

# puremacro: macroeconometrics and quantitative macroeconomics on the core scientific-Python stack

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## Summary

puremacro is a Python library for empirical macroeconomics and for solving quantitative macroeconomic models. It gives researchers, instructors and students one consistent interface to the field's standard methods, and a way to check the library's numbers on the machine that produces them.

On the empirical side it covers vector autoregressions (VARs) with recursive, long-run, sign, sign-zero, narrative and external-instrument identification ([Antolín-Díaz & Rubio-Ramírez, 2018](#); [Blanchard & Quah, 1989](#); [Mertens & Ravn, 2013](#); [Rubio-Ramírez et al., 2010](#); [Sims, 1980](#); [Stock & Watson, 2018](#); [Uhlig, 2005](#)); factor-augmented VARs ([Bernanke et al., 2005](#)); local projections ([Jordà, 2005](#)); ARCH, GARCH and GARCH-MIDAS volatility models ([Bollerslev, 1986](#); [Engle, 1982](#); [Engle et al., 2013](#)); autocorrelation-robust and weak-instrument-robust inference ([Anderson & Rubin, 1949](#); [Montiel Olea & Pflueger, 2013](#); [Newey & West, 1987](#)); dynamic panels ([Arellano & Bond, 1991](#)); difference-in-differences and synthetic control ([Abadie et al., 2010](#); [Arkhangelsky et al., 2021](#)); mixed-frequency nowcasting ([Bańbura & Modugno, 2014](#); [Giannone et al., 2008](#)); and text-based uncertainty indices ([Baker et al., 2016](#)). On the modelling side it solves dynamic programs by value-function iteration and endogenous grids ([Carroll, 2006](#); [Iskhakov et al., 2017](#)); heterogeneous-agent economies, including sequence-space HANK models ([Aiyagari, 1994](#); [Auclert et al., 2021](#); [Young, 2010](#)); continuous-time models ([Achdou et al., 2022](#)); projection methods on Chebyshev, finite-element and Smolyak grids ([Judd, 1992](#); [Krueger & Kubler, 2004](#); [McGrattan, 1996](#); [Smolyak, 1963](#)); linear rational-expectations (DSGE) models; quantitative spatial and trade models ([Allen & Arkolakis, 2014](#); [Caliendo & Parro, 2015](#)); and climate-economy models ([Golosov et al., 2014](#); [Nordhaus, 2018](#)).

Its estimators and solvers are written against NumPy ([Harris et al., 2020](#)), SciPy ([Virtanen et al., 2020](#)), pandas ([McKinney, 2010](#)) and Matplotlib ([Hunter, 2007](#)) alone, and the package contains no compiled code of its own. The same code therefore also runs in a web browser under Pyodide ([The Pyodide development team, 2021](#)), which ships those libraries compiled to WebAssembly.

## Statement of need

An applied macroeconomist's toolkit typically spans several ecosystems: reduced-form time series and panel econometrics from Python's statsmodels ([Seabold & Perktold, 2010](#)), arch ([Sheppard, 2025a](#)) and linearmodels ([Sheppard, 2025b](#)); structural identification from MATLAB toolboxes or R packages such as vars and lpirfs ([Adämmer, 2019](#); [Pfaff, 2008](#)); DSGE analysis in Dynare ([Adjemian et al., 2024](#)), which runs under MATLAB or GNU Octave; and heterogeneous-agent models in Python codes accelerated with Numba ([Auclert et al., 2021](#); [Batista et al., 2024](#); [Carroll et al., 2018](#)). Each tool is good at what it does, but combining them means several installations, often several languages, sometimes a commercial licence, and conventions that change from package to package. The cost is highest in teaching, where installation

42 problems consume class time and students' machines (managed laptops, low-end hardware,  
43 tablets) may not support the full stack.

44 puremacro is written for that audience: instructors, students and applied researchers in  
45 macroeconomics. It puts common estimators and models behind one set of conventions (shared  
46 arguments such as lags, horizon and ci, and immutable result objects with plotting and table  
47 export), keeps the dependencies of its numerical code to four ubiquitous libraries, and ships  
48 the evidence needed to trust its output: galleries of checks that users can run themselves.

## 49 State of the field

50 statsmodels, arch and linearmodels are the reference Python implementations of reduced-form  
51 time series, conditional volatility and panel and instrumental-variable estimation. puremacro  
52 does not aim to replace them; it uses them as test oracles (see *Software design*). They do not,  
53 however, implement the identification schemes that define structural macroeconometrics (sign,  
54 sign-zero, narrative and external-instrument restrictions), nor do they solve heterogeneous-agent  
55 or general-equilibrium models. For models, widely used tools are Dynare, the sequence-space  
56 Jacobian toolkit, HARK and QuantEcon.py. In their domains they are more mature than  
57 puremacro and, for large heterogeneous-agent problems, faster: puremacro trades speed for  
58 portability.

59 That portability can be stated precisely. As of September 2026 the Pyodide distribution  
60 includes NumPy, SciPy, pandas, Matplotlib and statsmodels, but not arch, linearmodels or  
61 Numba; arch and linearmodels publish no pure-Python wheels that a browser could install  
62 at run time; and HARK and QuantEcon.py declare Numba as a required dependency, while  
63 the sequence-space Jacobian toolkit uses it in its core modules. In a browser, the volatility,  
64 panel-IV and heterogeneous-agent layers of the usual stack are therefore unavailable, and  
65 Dynare needs a different language runtime altogether.

66 This also answers the build-versus-contribute question. Individual estimators could be  
67 contributed to existing packages, and some would be welcome there. What cannot be  
68 contributed is the property that makes the collection usable on a constrained machine: one  
69 import surface whose numerical code depends on nothing beyond four core libraries. That is  
70 an invariant of the whole library, enforced by tests, not a feature that can be added to another  
71 project's dependency graph.

## 72 Software design

73 **An import invariant.** Library modules import NumPy, SciPy, pandas and Matplotlib (plus  
74 requests, in the data layer) at module scope; anything else, from Parquet support to optional  
75 accelerators, is optional and imported only when available. Two tests enforce the rule against  
76 the packages most likely to leak in. The first imports each of the 597 library modules (examples,  
77 teaching helpers, text-scraping sources and optional Numba kernels are excluded) and fails  
78 if statsmodels, linearmodels, arch or the scraping packages bs4, pdfplumber and pypdf have  
79 entered sys.modules. The second repeats the sweep in a subprocess in which those packages  
80 cannot be imported, so a module that would break on a machine without them fails in  
81 continuous integration (CI) on a machine that has them. The wheel is pure Python, and its  
82 base install adds only requests to the four core libraries; the Parquet and Excel engines are an  
83 optional extra.

84 **Oracles that do not ship.** The packages forbidden at run time are the test suite's references.  
85 Scripts in the repository run statsmodels, arch, linearmodels and esda once on fixed inputs  
86 and store their outputs as package data, so the validation gallery can compare against them  
87 without importing them; an opt-in test marker recomputes the stored outputs with the installed

88 packages to detect drift. The reference implementations thus certify the library without  
89 becoming its dependencies, which is what lets breadth and portability coexist.

90 **Degrading rather than failing.** Where an external tool does better, puremacro uses it if present:  
91 seasonal adjustment calls X-13ARIMA-SEATS through statsmodels when both are installed  
92 and otherwise falls back to a native X-11. A runtime module detects the host (CPython  
93 or Pyodide), the device class and the four capabilities that differ away from a workstation  
94 (sockets, Parquet, threads and a writable filesystem), and offers opt-in adaptations such as a  
95 browser fetch transport. A few dynamic-programming and projection solvers have optional  
96 Numba kernels and experimental MLX and CuPy paths; NumPy remains the reference path  
97 and the one tested in CI.

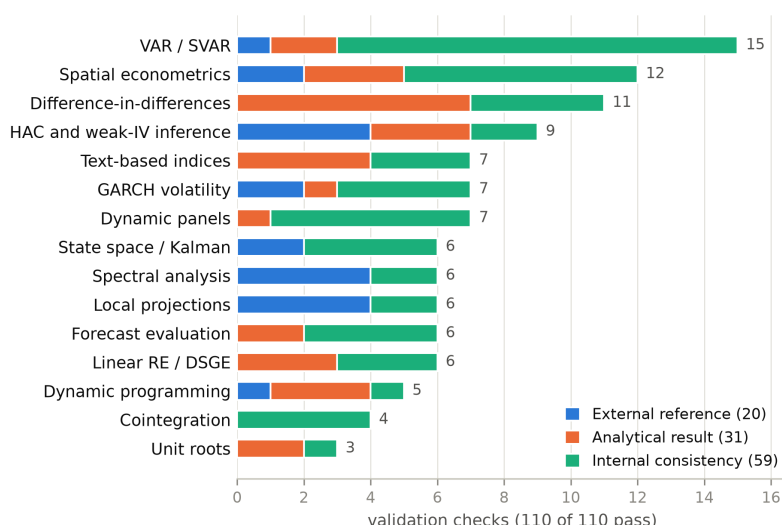
98 **Costs.** Writing everything in vectorised NumPy makes large heterogeneous-agent problems  
99 slower than in JIT-compiled toolkits, and derivatives that other libraries obtain by automatic  
100 differentiation must be coded by hand. The browser is a best-effort target rather than a  
101 supported one: Parquet and Excel files need engines that not every Pyodide distribution  
102 provides. A headless harness in the repository runs the library in Node.js against a pinned  
103 Pyodide, and an opt-in release gate installs the package there exactly as the playground does  
104 and runs a 31-test smoke subset of the suite.

105 **Scale.** Version 4.4.0 comprises about 810 modules and 281,000 lines of Python, exercised  
106 by about 18,800 tests that CI runs on Linux, macOS and Windows under Python 3.11–3.13.  
107 Before a release is tagged, a script checks the suite against a recorded baseline, the import  
108 invariant, a snapshot of the public API that must be regenerated deliberately when the interface  
109 changes, that every shipped file still parses on the oldest supported Python, and that the  
110 version string agrees across the package metadata, changelog and citation file.

## 111 Verification

112 The validation gallery, `puremacro.validation.scorecard()`, runs 110 checks across 15  
113 subsystems in under a minute, with none of the oracle packages installed; all pass (Figure 1).  
114 The checks differ in strength, and each records its reference. Twenty compare against an  
115 external reference: stored outputs of statsmodels, arch, linearmodels and esda, SciPy results  
116 computed at run time, or a published table. Agreement with independent implementations  
117 is typically at machine precision (median relative difference of order  $10^{-15}$ ); the exception is  
118 GARCH, whose estimates differ from arch's by up to 0.35% because the two packages use  
119 different optimisers. Thirty-one checks compare against analytical results, and 59 test internal  
120 consistency, such as agreement between alternative algorithms and recovery of parameters  
121 from simulated data.

122 A separate replication gallery, `puremacro.replication.scorecard()`, reproduces published  
123 estimates of the return to schooling (Card, 1995), a textbook logit of married women's labour-  
124 force participation on Mroz's (1987) data, and the output effect of tax changes (Romer &  
125 Romer, 2010), as well as two qualitative predictions of incomplete-markets models (Aiyagari,  
126 1994; Huggett, 1993): precautionary saving holds the interest rate below the rate of time  
127 preference, and the rate falls as income risk rises. Most model solvers (sequence-space,  
128 continuous-time, spatial, trade and climate) are covered by unit tests but not yet by gallery  
129 checks.



**Figure 1:** The validation gallery of puremacro 4.4.0: 110 checks in 15 subsystems, by kind of reference; all pass. *External reference*: stored outputs of statsmodels, arch, linearmodels or esda, SciPy computed at run time, or a published table. *Analytical result*: a closed form, or an effect planted in simulated data. *Internal consistency*: agreement between alternative algorithms, identities that correct output must satisfy, or recovery of parameters from simulated data. Regenerate with python paper/make\_scorecard\_fig.py.

## Research impact statement

puremacro is young (public since July 2026) and has not yet been used in published research or by other packages. Its significance is near-term, and rests on evidence a reviewer can check.

*Teaching.* The library was written for, and is used in, the author's course *Macroeconomía Avanzada* at ITAM, whose Fall 2026 edition builds its computational material on it. The course's 22 Spanish lesson notebooks are in the repository (notebooks/course), and 20 of them call puremacro directly.

*Reproducible material.* The repository contains 70 bilingual (English/Spanish) pairs of worked-example notebooks; all of them, with the lesson notebooks, are also published as a JupyterLite ([JupyterLite contributors, 2021](#)) site that runs them in the browser.

*Community readiness.* The package is released on PyPI under the MIT licence, with documentation at <https://jalonso1979.github.io/puremacro/>, contribution guidelines, and CI on three operating systems.

## AI usage disclosure

Generative AI was used extensively in writing puremacro, its documentation and this paper. Anthropic's Claude models, used through the Claude Code agent (Claude Opus 5, Claude Fable 5 and Claude Fable 5.1), generated or co-wrote much of the code, tests, documentation and notebooks: 171 of the 310 commits in the public repository carry a Claude co-author trailer. Google's Jules coding agent contributed 20 pull requests (refactoring, performance and test improvements), each reviewed and merged by the author. Claude also drafted and revised this paper, including checking its claims against the code and its references against Crossref.

The author set the scope and made the core design decisions (the import invariant, validation against stored outputs of reference implementations, and the shared interface conventions) and reviewed, edited and validated all AI-assisted output before it was merged. Correctness does

not rest on that review alone: it is checked by the test suite, the validation and replication galleries, and CI.

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